

# Lesson 04 Practice: Answers and Rationales

## JKSSB Computer Awareness

October 10, 2026

**Note:** All 25 items are original exam-pattern questions (the *[Original]* items plus the extension items tagged *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]* and *[JKSSB-pattern]*, including MSQs); the correct letter and the reason each other option is wrong are given below.

- Q1. A.** ALU stands for Arithmetic Logic Unit. “Application Logic Unit”, “Arithmetic Language Unit” and “Advanced Logic Unit” are invented expansions that do not name the unit correctly.
- Q2. C.** The ALU performs arithmetic and logical (comparison) operations. Permanent storage (A) is a storage job, control of I/O devices (B) is the control unit’s work, and supplying power (D) is the power supply’s role.
- Q3. D.** The control unit directs and coordinates all other units, including the ALU. Options A, B and C name units that are themselves coordinated, not the coordinating unit.
- Q4. C.** The CPU consists of the ALU, the control unit and registers. Input/output units and memory (A) are separate units; main and secondary storage (B) are storage, not the CPU; monitor, keyboard and mouse (D) are peripheral devices.
- Q5. A.** CPU expands to Central Processing Unit. The other three options are incorrect wordings that are not the standard expansion.
- Q6. A.** Registers are small, fast storage inside the CPU used to hold data and addresses during processing. Secondary storage (B) and a hard disk (C) are slower and outside the CPU; the output unit (D) presents results.
- Q7. B.** The address bus carries the address of a memory location from the CPU to memory. Data travel on the data bus (A), control signals on the control bus (C), and power is supplied separately (D), not over the address bus.
- Q8. B.** The data bus is bidirectional, moving data and instructions both ways. The address bus (A) is one-way from the CPU, the control bus (C) carries control signals, and there is no separate “power bus” in this three-bus model (D).
- Q9. C.** The control bus carries control and timing signals such as read and write. Addresses travel on the address bus (A), data on the data bus (B), and keyboard input is not what the control bus carries (D).
- Q10. C.** This is the incorrect statement: the CPU accesses main memory directly, not secondary storage. Statements A, B and D are all true of the ALU, the control unit and the CPU respectively.

- Q11. B.** The CPU directly accesses main memory (primary storage). Secondary storage (A), a pen drive (C) and a hard disk (D) are all secondary storage, reached only through main memory and the OS.
- Q12. C.** The hard disk is secondary storage. RAM (A), ROM (B) and registers (D) are primary storage or CPU storage, not secondary storage.
- Q13. A.** Main memory holds the data and instructions currently being processed. Option B is the classic error (the CPU does not process directly from secondary storage); RAM is volatile, not non-volatile secondary storage (C); and registers are faster than secondary storage (D).
- Q14. D.** The instruction cycle runs fetch, decode, execute. Options A, B and C scramble this fixed order.
- Q15. C.** During fetch the CPU obtains the next instruction from main memory. Arithmetic is the execute step (A); writing the result back is the store step (B); displaying output (D) is not the fetch step.
- Q16. C.** The stored-program concept stores both instructions and data in the same main memory. Option A wrongly excludes instructions; option B wrongly limits programs to secondary storage; option D places instructions in an output device.
- Q17. A.** Storing instructions and data together in memory is the von Neumann architecture idea. The address bus (B), secondary storage (C) and the output unit (D) are unrelated to the concept.
- Q18. D.** All three statements are correct: the address bus carries the address from the CPU to memory, the data bus is bidirectional, and the control bus carries control signals. Options A, B and C each omit at least one true statement.
- Q19. D.** Expansion slots let you add cards such as graphics or network cards. They do not permanently store the OS (A), replace the control unit (B), or supply power (C).
- Q20. C.** The control bus carries control signals. The address bus is unidirectional, not bidirectional (A); the data bus is bidirectional, not unidirectional (B); and the data bus carries data, not addresses (D).
- Q21. A.** The CPU processes instructions and is called the brain. B, C, and D are peripherals or storage, not the processing brain.
- Q22. A.** The ALU handles arithmetic and logical work. B, C, and D describe peripheral or unrelated roles.
- Q23. A.** CU is the Control Unit. B, C, and D are not the recognised expansion.
- Q24. A, B, C.** The ALU, CU, and registers form the CPU. The hard disk (D) is secondary storage outside the CPU.
- Q25. A, B, D.** The CPU composition, the data-flow order, and memory's role are correct. C is false: the CU directs operations; it does not permanently store files.
- Q26. B.** The Control Unit sequences operations and directs data flow. The ALU performs arithmetic/logic; memory stores data; input supplies data. The cited key is provisional.

- Q27. B.** The ALU performs arithmetic and logical operations. The Control Unit directs execution, the Program Counter holds an instruction address, and the input unit supplies data.
- Q28. A.** The address bus carries location addresses. The data bus carries data; the control bus carries timing/control signals; expansion buses connect peripherals.
- Q29. B.** Input supplies data, processing transforms it, and output presents results. The other sequences reverse that basic order.
- Q30. A, B, C, D.** A general-purpose computer has input, memory, processing (CPU), and output functions.

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